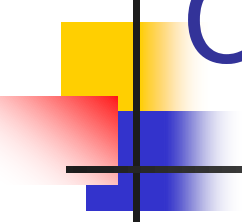


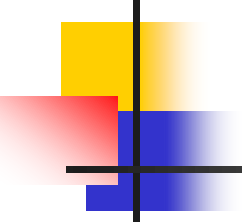
Lecture 1

Introduction of Communications



Communication Systems

- **Communication systems** are designed to transmit information from the information source (voice, video, data, etc.) to the information user through a communication channel
- **Information source** is a device that produces the information to be communicated by the communication system.
- **Analog information source** has output that assumes values in a continuous range of amplitudes and time e.g speech signal, music signal, and television signal, are analog information sources.

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- **Discrete information sources** output takes a value from a finite set of possible values e.g. computer data, text, etc



Communication Systems Mode

- Two basic modes of communications
 - Broadcasting
 - Single powerful transmitter and many receivers
 - TV, Radio, etc.
 - Point-to-point communication
 - Link between a single transmitter and a receiver
 - Telephone



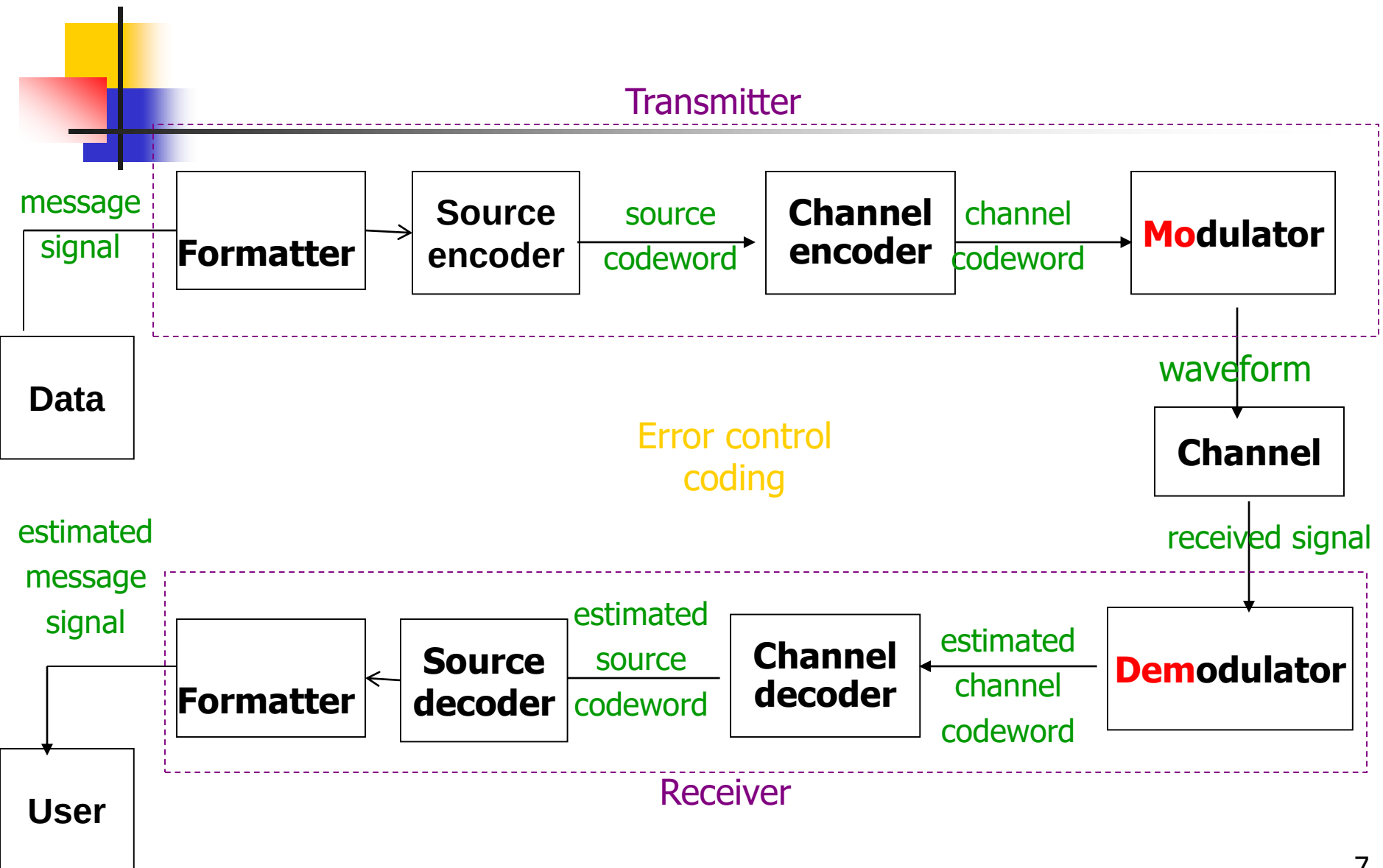
Communication Resources

- Two primary resources
 - Transmitted power
 - Average power of transmitted signal
 - Channel bandwidth
 - Band of frequencies allocated for transmission
- System design objective
 - Use two resources as efficiently as possible.
 - Power limited vs. Band limited



Elements of Communication System

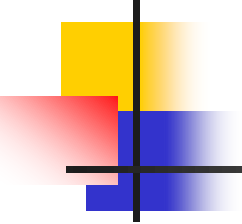
- Three basic elements of a communication system are
 - i. Transmitter
 - ii. Channel
 - iii. Receiver



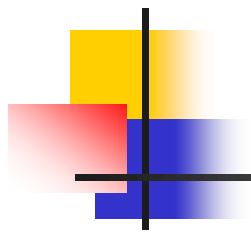


Transmitter

- A **transmitter** converts the message signal from the information source into a form suitable for transmission over the communication channel
- Typical processes at the transmitter include formatting, source and channel coding, multiplexing, modulation, frequency spreading, etc

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- **Formatting** transforms analog information from the information source to digital form (usually binary digits “bits”)
 - Examples are pulse code modulation (PCM) which involves sampling, quantization and encoding
 - **Formatting**, pulse modulation and band-pass digital modulation are essential for any digital communication system

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- **Source Encoding** reduces redundancy in binary representation of source information for increased efficiency in utilization of communication channel bandwidth (for transmission) or memory (for storage)
 - **Encryption** prevents unauthorized persons from extracting/injecting information from/to the channel to improve communication privacy and authentication
 - **Channel Encoding** adds controlled redundant bits to the information bits to form a sequence of channel symbols for detection and correction of errors caused by channel impairments for improving system reliability and efficiency



- **Multiplexing and Multiple Accessing** combine information signals from different sources to share a common communication resource, e.g., frequency division multiple access (FDMA), time division multiplexing (TDMA), etc .
- **Pulse Modulation** transforms a binary representation of information to a baseband waveform



Communication Channel

- A **communication channel** is a physical medium that connects the transmitter and the receiver
- Physical channels include a pair of wires, an optical fiber, free space, etc
- Effects of channels on a transmitted signal include attenuation, amplitude and phase distortion, fading and corruption by additive noise and interference.



Channel Capacity

- The most important question for a communication channel is the maximum rate at which it can transfer information.
- There is a theoretical maximum rate at which information passes error free over the channel, called the channel capacity **C**.
- The famous **Hartley-Shannon Theory** states that the channel capacity **C** is given by:

$$C = B \cdot \log(1 + (S/N)) \text{ b/s}$$

where B is the bandwidth, S/N is the signal-to-noise ratio.



Channel Capacity

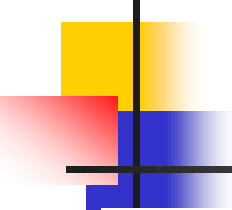
- AWGN channel

$$C = W \log_2 (1 + \text{SNR}) \text{ bit/sec},$$

where W denotes the channel bandwidth and SNR denotes the signal to noise ratio.

- For an AWGN, SNR can be expressed as transmit power P and noise power spectral density N_o watt/Hz , the capacity C is given by the formula

$$C = W \log_2 \left(1 + \frac{P}{N_o W}\right) \text{ bit/sec}$$



For energy per information bit E_b , the transmitted power $P = E_b R$ and $R < C$ implies that

$$R < W \log_2 \left(1 + \frac{E_b}{N_o} \frac{R}{W} \right) \text{ bit/s}$$

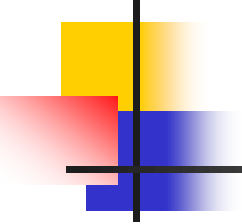
$$r < \log_2 \left(1 + \frac{E_b}{N_o} r \right) \text{ bit/s}$$

$$\frac{E_b}{N_o} > \frac{2^r - 1}{r} \text{ where } r = \frac{R}{W} \text{ is the spectral efficiency}$$

Shannon Capacity

Shannon Theory

- It establishes that given a noisy channel with information capacity C and information transmitted at a rate R , then if $R < C$, there exists a coding technique which allows the probability of error at the receiver to be made arbitrarily small. This means that theoretically, it is possible to transmit information without error up to a limit, C .
- The converse is also important. If $R > C$, the probability of error at the receiver increases without bound as the rate is increased. So no useful information can be transmitted beyond the channel capacity. The theorem does not address the rare situation in which rate and capacity are equal.



Example

- Suppose the spectrum of a channel is between 3 MHz and 4 MHz, and $\text{SNR}_{\text{dB}} = 24\text{dB}$. Find the channel capacity

So,

$$B = 4 \text{ MHz} - 3 \text{ MHz} = 1 \text{ MHz}$$

$$\text{SNR}_{\text{dB}} = 24 \text{ dB} = 10 \log_{10}(\text{SNR}) \rightarrow \text{SRN} = 251$$

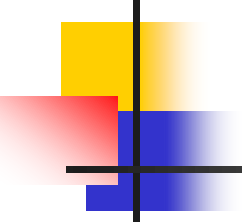
- Using Shannon's formula, the capacity limit C is:

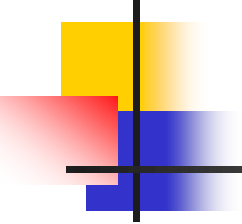
$$C = 10^6 \times \log_2(1+251) \approx 8 \text{ Mbps.}$$

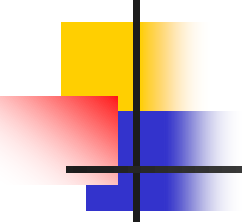
- **Twisted-pair wire-line channel:**

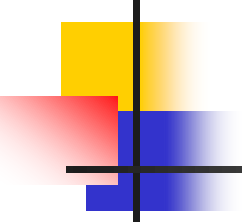
Bandwidth-limited with several hundred kilohertz (kHz).

- Transmission rate up to 16.8 kb/s.
Susceptible to electromagnetic interference (EMI).

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- **Coaxial cable channel:** Higher bandwidth - several MHz. High immunity to EMI.
 - Transmission rate up to 274 Mb/s. Used in LAN and in cable television systems

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- **Fiber Optic cable channel:** Transports information in a form of light signals.
 - Has enormous potential bandwidth ($\approx 2 \times 10^{13}$), low signal attenuation (≈ 0.1 dB/km)
 - Has high immunity to EMI, small size and weight, etc.

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- **Wireless broadcast channels:**
Supports the transmission of radio and television signals.
 - The transmission and reception is facilitated by antennas

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- **Mobile radio channels:** Introduce mobility into a public telecommunication network. Mobile radio channels are affected by multi-path phenomena
 - **Satellite channels:** Provide broad-area coverage, reliable transmission links, and wide transmission bandwidths
 - A message signal is transmitted from the Earth station to a satellite via an uplink.

- **Satellite channels:**

A signal is amplified in a transponder and then retransmitted from the satellite to another Earth station via a downlink

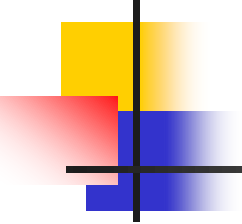
- **Communication Channels Modeling:**

Characterizing Effects of communication channels on transmitted message signals and mitigation methods are analyzed by using channel mathematical models



Receiver

- The receiver reverses the signal processing transformation performed by the transmitter.
- The combined process of modulation and demodulation is called MODEM
- Typical receiver processes are demodulation, demultiplexing, channel decoding, decrypting, source decoding and formatting.

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- Performance of a digital communication system receiver is measured by the probability of incorrectly detection a digit, or the probability of error.
 - Probability of error P_e depends on
 - Channel coding characteristics
 - ✓ Type of the waveforms used to transmit the data, which determines the transmission bandwidth .
 - ✓ The transmit power
 - ✓ Channel characteristics
 - ✓ Demodulation Method
 - ✓ Decoding Method



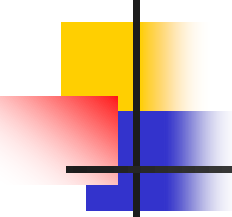
Modulation

- Modulation:
 - Modifies the message signal into a form suitable for transmission over the channel.
- Demodulation:
 - Recreates the original message signal from a degraded version of the transmitted signal after propagation through the channel.
 - Due to the presence of noise, the original message signal cannot be recreated exactly.
 - The degradation is influenced by the type of modulation scheme.



Modulation

- Continuous wave (CW) modulation
 - Carrier is a sinusoidal wave.
 - Amplitude modulation (AM), frequency modulation (FM), phase modulation (PM)
- Pulse modulation
 - Carrier is a periodic sequence of rectangular pulses.
 - Pulse-amplitude modulation (PAM) or amplitude-shift keying (ASK), pulse-duration modulation (PDM), pulse-position modulation (PPM)
- Other names: frequency shift keying (FSK), amplitude-shift keying (ASK), on-off keying (OOK), phase-shift keying (PSK), M-ary orthogonal keying, etc.



Multiplexing

- Combines several message signals for their simultaneous transmission over the same channel.
 - Frequency-division multiplexing (FDM)
 - CW modulation is used.
 - Assigns message signal distinct carrier frequency.
 - Time-division multiplexing (TDM)
 - Pulse modulation is used.
 - Different time slots
 - Code-division multiplexing (CDM)
 - Each message is identified by a distinctive code.
 - Message signals are permitted to overlap in both time and frequency.



Error Control Coding

- Channel encoder produces a new sequence of symbols called the channel codeword.
- Controlled redundancy exists in the construction of channel codeword.
 - Channel codeword is longer than source codeword.
- Benefits
 - In principle:
 - If you transmit information at a rate $R < C$, then the error-free transmission is possible.
 - In practice:
 - Reduce the error rates
 - Reduce the transmitted power requirements



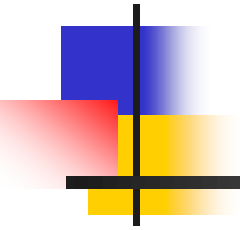
Networks: key issues

- Network criteria
 - Performance
 - Throughput
 - Delay
 - Reliability
 - Data transmitted are identical to data received.
 - Measured by the frequency of failure
 - The time it takes a link to recover from a failure
 - Security
 - Protecting data from unauthorized access



Terminology

- The *throughput* or *bandwidth* of a channel is the number of bits it can transfer per second
- The *latency* or *delay* of a channel is the time that elapses between sending information and the earliest possible reception of it



Thank You!